

Origins Data Science Lab

Introduction to Machine Learning

Written Exam

Date: 9th April 2026

Time: 10:00–12:00

Duration: 1 hour 30 mins

Instructions:

- Answer all questions. Read the question carefully.
- Show all working for full credit.
- Write clearly and justify your reasoning.
- Ask for more paper if you need it, *labelling each sheet with your name and student ID*.
- Place your student ID in clear view on your desk.
- Do not remove this exam sheet from the lecture theatre.

Question 1

[6 marks]

Explain the bias-variance tradeoff in machine learning. You may refer to Figure 1 in your explanation.

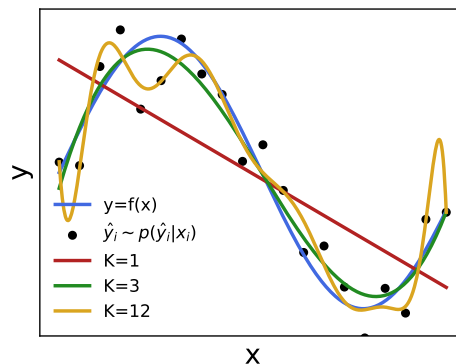


Figure 1: Diagram showing linear regression fits $\hat{f}(x) = \sum_{k=1}^K \theta_k x^k$ with increasing degree in the polynomial basis K used to fit the data.

Question 1 (continued)

Question 2**[14 marks]**

Let T be an estimator of a parameter θ such that $\mathbb{E}[T] = \theta$.

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable, non-linear function, and define a new estimator

$$\hat{\phi} = g(T).$$

- (a) Show that, in general,

$$\mathbb{E}[g(T)] \neq g(\theta),$$

by performing a second-order Taylor expansion of $g(T)$ around θ .

- (b) Then derive the leading-order correction term in $\mathbb{E}[g(T)]$ in terms of derivatives of g and moments of T , expressing your result where appropriate in terms of $\text{Var}(T)$.

Recall that $\text{Var}(T) = \mathbb{E}[(T - \theta)^2]$.

- (c) Hence, show that $g(T)$ is not, in general, an unbiased estimator of $g(\theta)$.
- (d) Under what condition on g would $g(T)$ remain an unbiased estimator of $g(\theta)$ for any unbiased estimator T , regardless of the distribution of T ?

Question 2 (continued)

Question 2 (continued)

Question 3**[20 marks]**

Let $x \in \mathbb{R}^n$ be a data vector distributed as

$$x \sim \mathcal{G}[x \mid A\theta, \Sigma],$$

where A is a known $n \times p$ matrix, $\theta \in \mathbb{R}^p$ is an unknown parameter vector, and Σ is a known positive-definite covariance matrix.

- (a) Write down the log-likelihood function $\log p(x \mid \theta)$, ignoring additive constants independent of θ .
- (b) Show that the maximum likelihood estimator $\hat{\theta}$ satisfies

$$A^T \Sigma^{-1} A \hat{\theta} = A^T \Sigma^{-1} x.$$

You may define $v(\theta) = x - A\theta$, and use that

$$\nabla_{\theta}(v^T \Sigma^{-1} v) = -2A^T \Sigma^{-1}(x - A\theta).$$

- (c) Hence write down an explicit expression for $\hat{\theta}$.
- (d) Show that $\hat{\theta}$ is an unbiased estimator of θ .
- (e) Compute the Fisher information matrix for this model:

$$F(\theta) = \mathbb{E}_x[s(\theta)s(\theta)^T], \quad s(\theta) = \nabla_{\theta} \log p(x|\theta).$$

You may use that $\mathbb{E}[x] = A\theta$.

You may also express the score $s(\theta)$ in terms of $x - A\theta$.

Question 3 (continued)

Question 3 (continued)

Question 4**[8 marks]**

Consider a regression problem where the target y is modelled as a linear combination of basis functions of the input x .

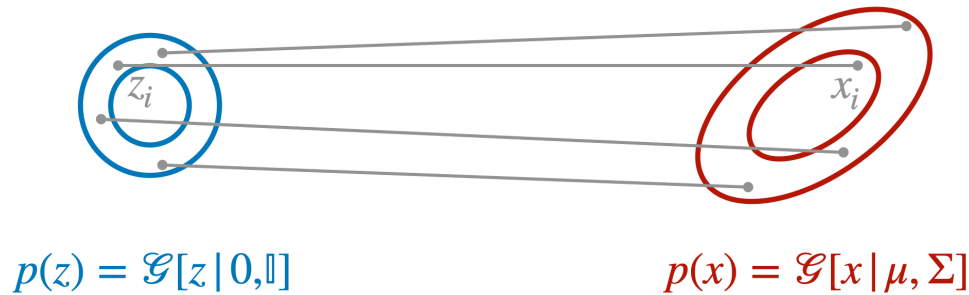
- (a) Explain the benefit of representing the model using a fixed polynomial basis of order K , i.e. $\phi(x) = (1, x, x^2, \dots, x^K)$, compared to using only linear functions of the input.
- (b) Define the design matrix X for this model, clearly stating its dimensions and entries.
- (c) Write down the likelihood function that we maximise with respect to the parameters θ to obtain the estimator $\hat{\theta}$.
- (d) Briefly explain how neural networks can be viewed as learning a data-driven basis, and how this can improve flexibility compared to fixed basis functions.

Question 4 (continued)

Question 4 (continued)

Question 5

[10 marks]

Figure 2: Diagram showing a change of variables between two distributions, $p(z)$ and $p(x)$.

Consider a normalising flow defined by an affine transformation

$$x = f(z) = Az + b,$$

where $z \in \mathbb{R}^n$, A is an invertible $n \times n$ matrix, and $b \in \mathbb{R}^n$.

Assume the latent variable is distributed as a standard normal,

$$z \sim \mathcal{G}[z | 0, I].$$

where I is a n -dimensional identity matrix.

- (a) Using the change of variables formula, write down the probability density $p(x)$ induced by this transformation.
- (b) Show that x is distributed as a Gaussian, and determine its mean and covariance in terms of A and b .
- (c) Suppose we wish to model a Gaussian distribution

$$x \sim \mathcal{G}[x | \mu, \Sigma].$$

Find A and b such that the flow exactly maps the latent distribution to this target distribution.

- (d) Briefly explain why affine normalising flows are sufficient to represent Gaussian distributions, but insufficient for modelling more complex distributions.

Question 5 (continued)

Question 5 (continued)